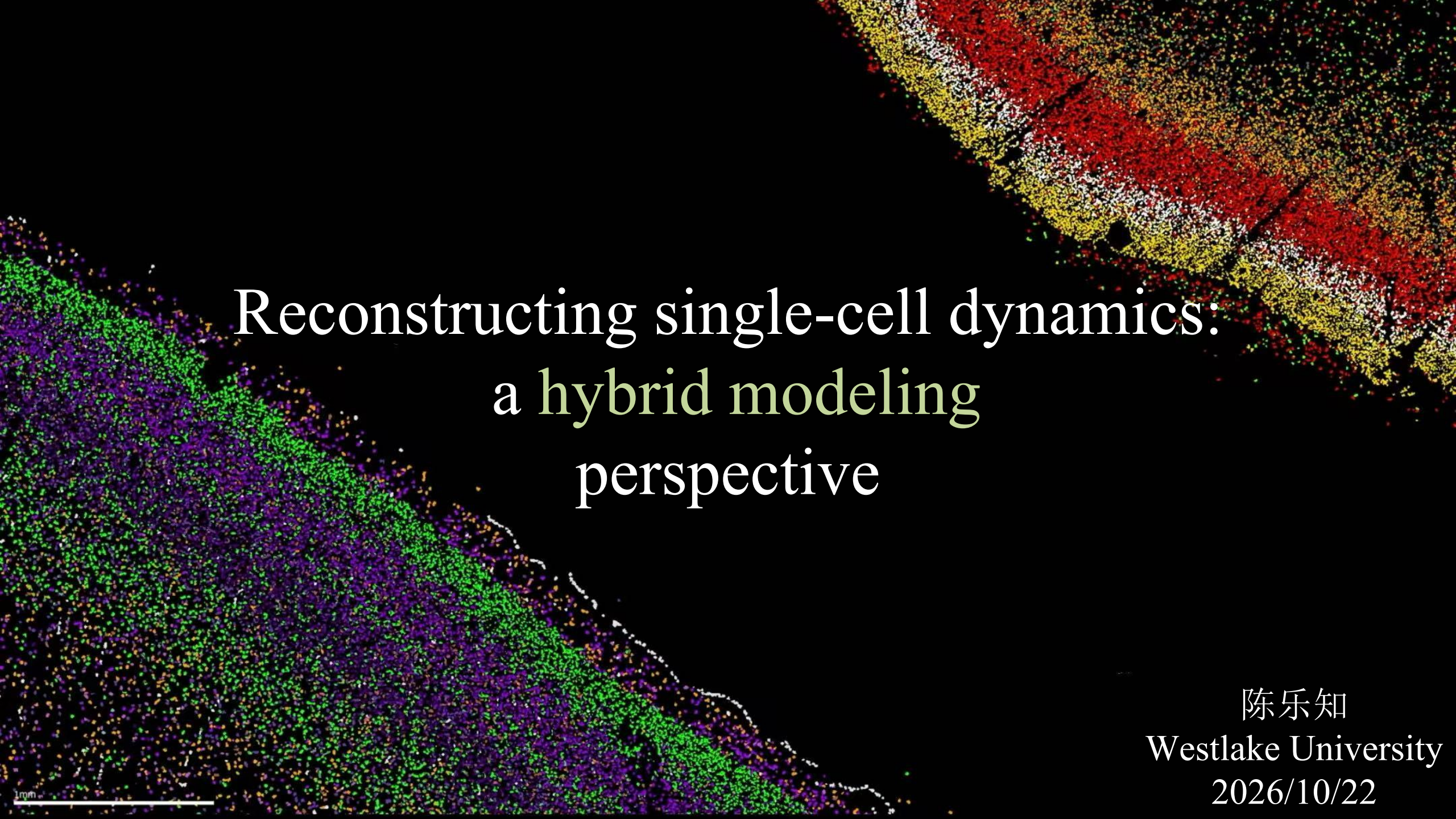
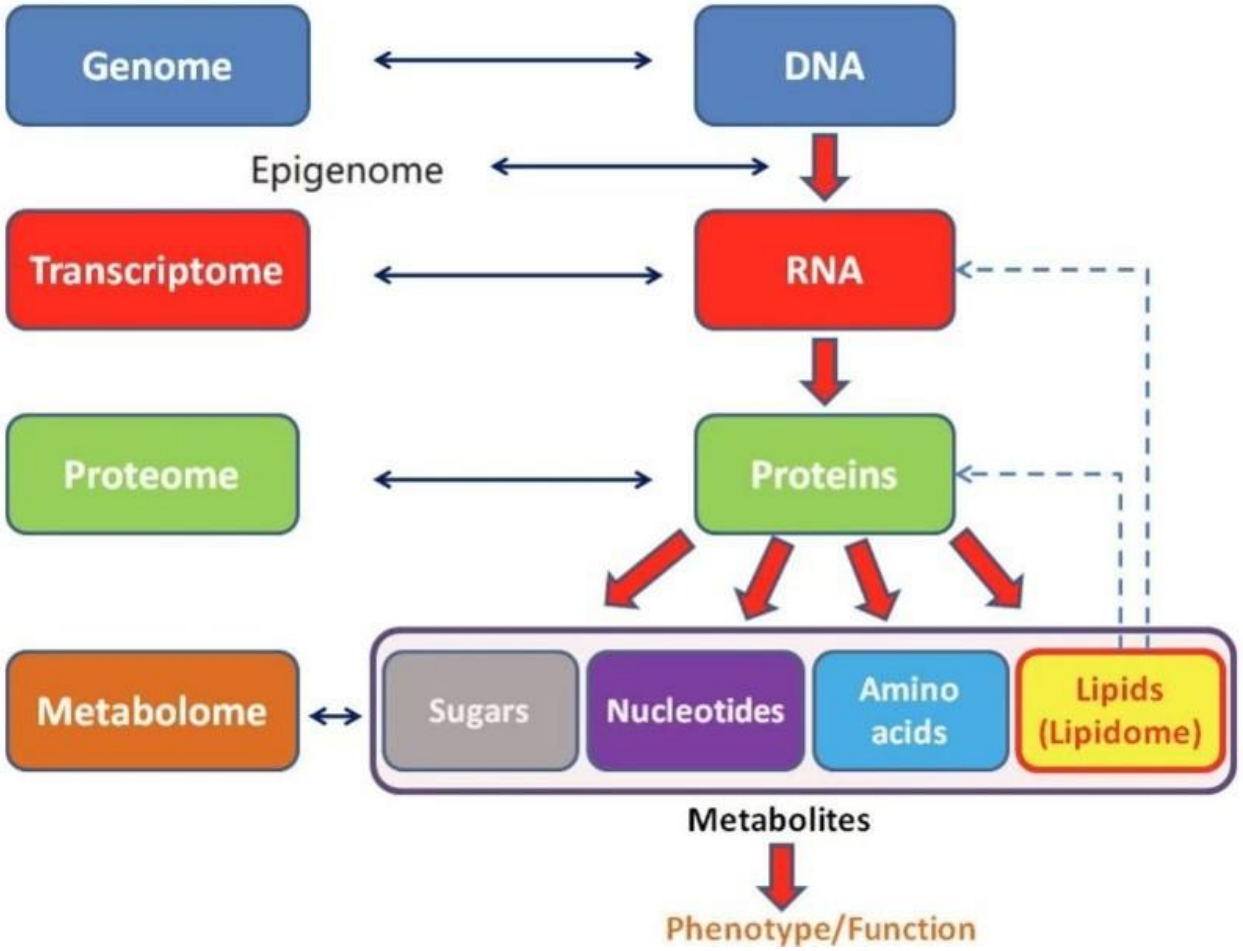


Reconstructing single-cell dynamics: a hybrid modeling perspective

陈乐知
Westlake University
2026/10/22

Modeling living systems

mechanistic modeling v.s. data-driven modeling

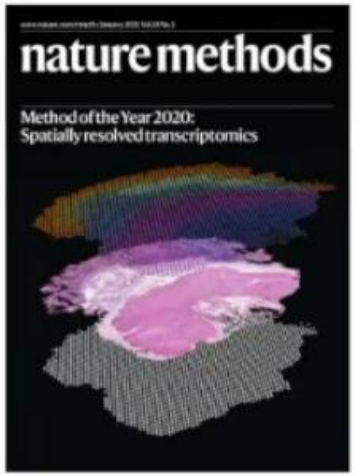


$$\frac{\partial \rho}{\partial t} = D_\rho \nabla^2 \rho - \chi \nabla \cdot (\rho \nabla c)$$

$$\frac{\partial c}{\partial t} = D_c \nabla^2 c + \alpha \rho - \beta c$$

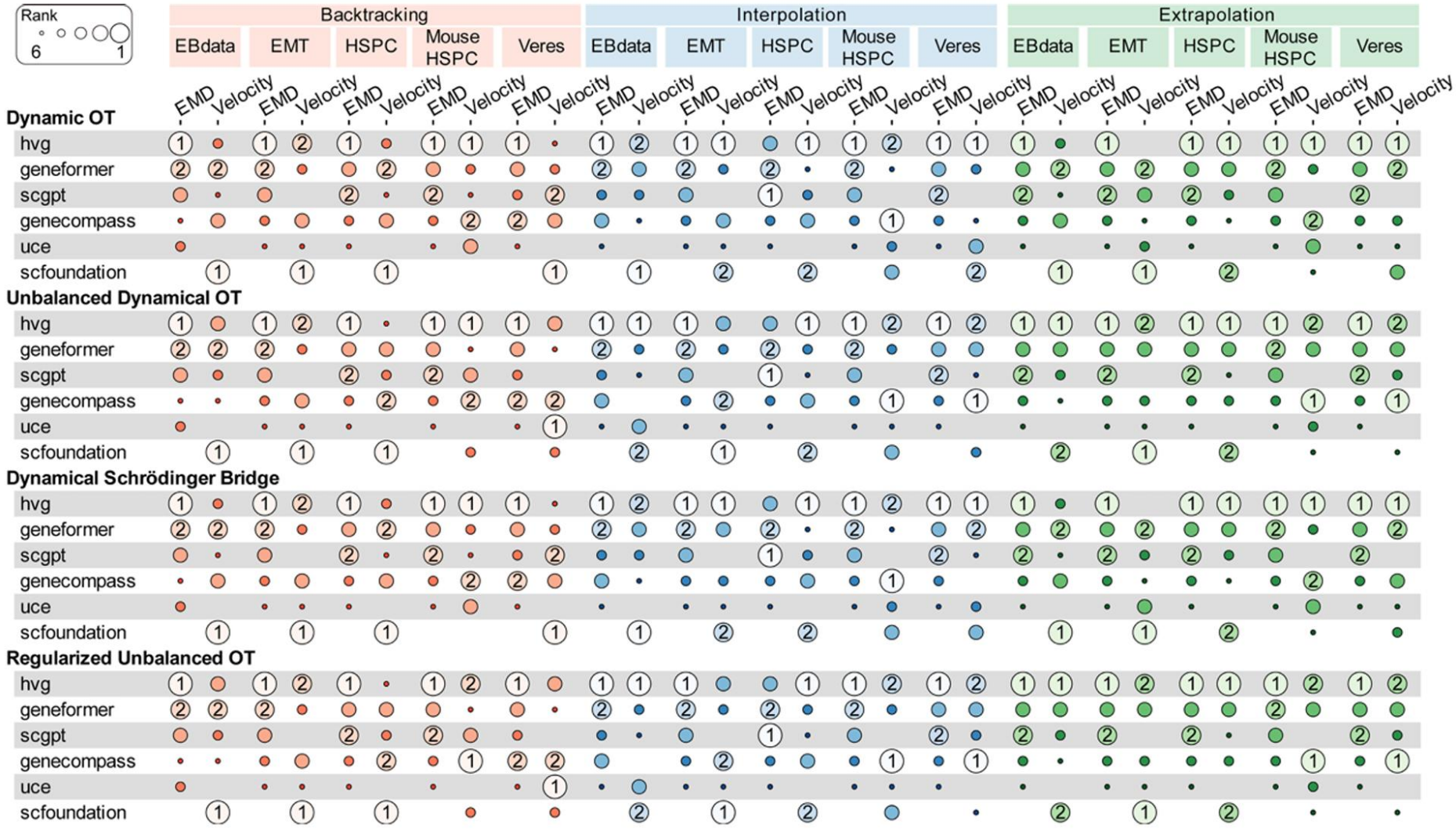


scRNA-seq (2018)



Spatial transcriptomics (2020)

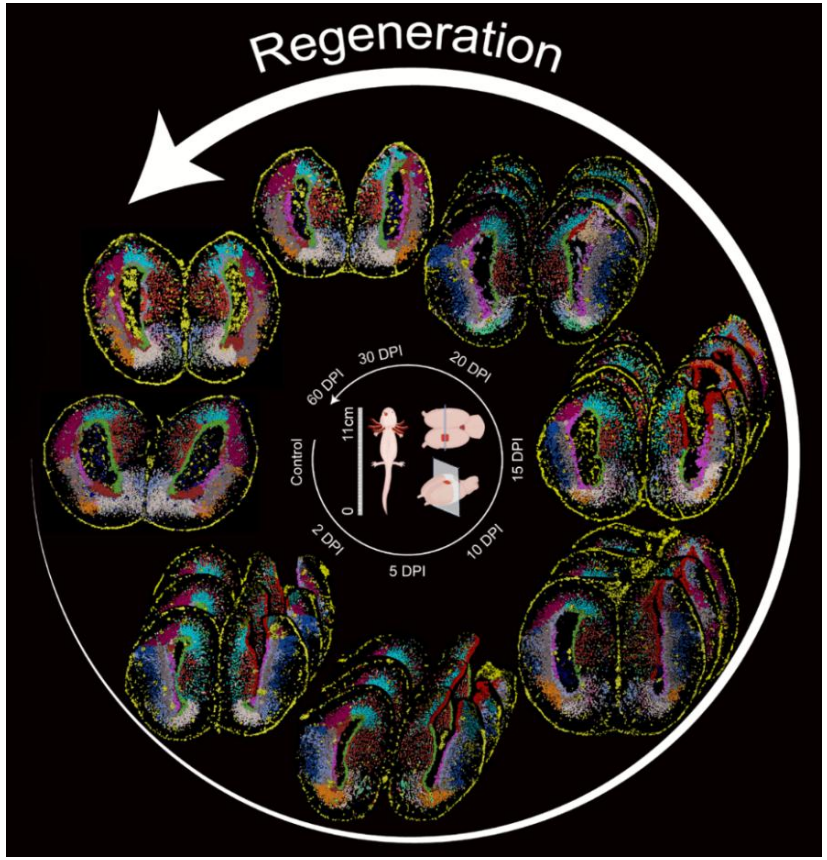
But data-driven methods do not always work !



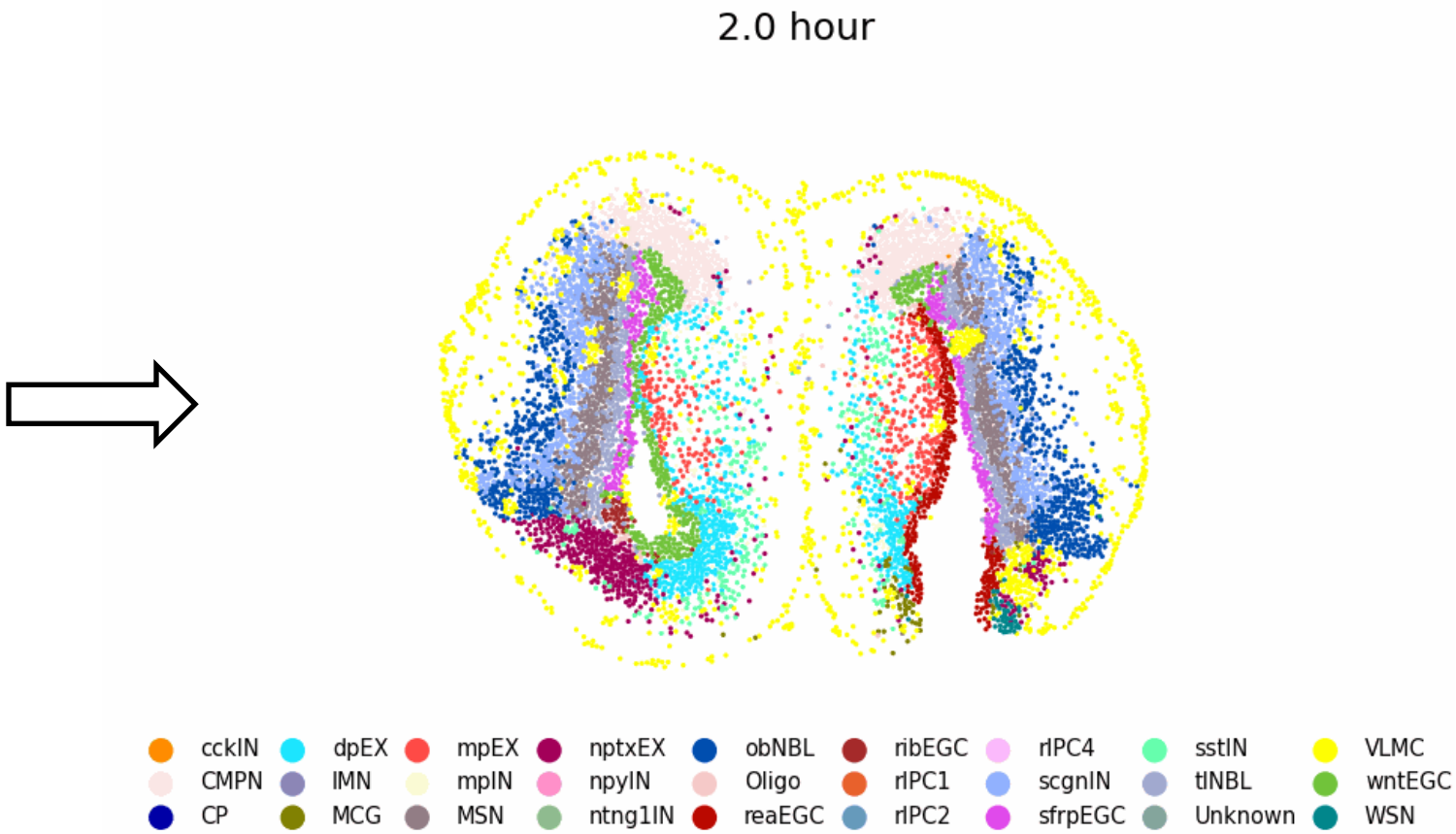
Zhou, et al. bioRxiv 2026

Why not combine the strengths of both ? That's exactly what hybrid modeling is about !

Single-cell spatiotemporal dynamics reconstruction

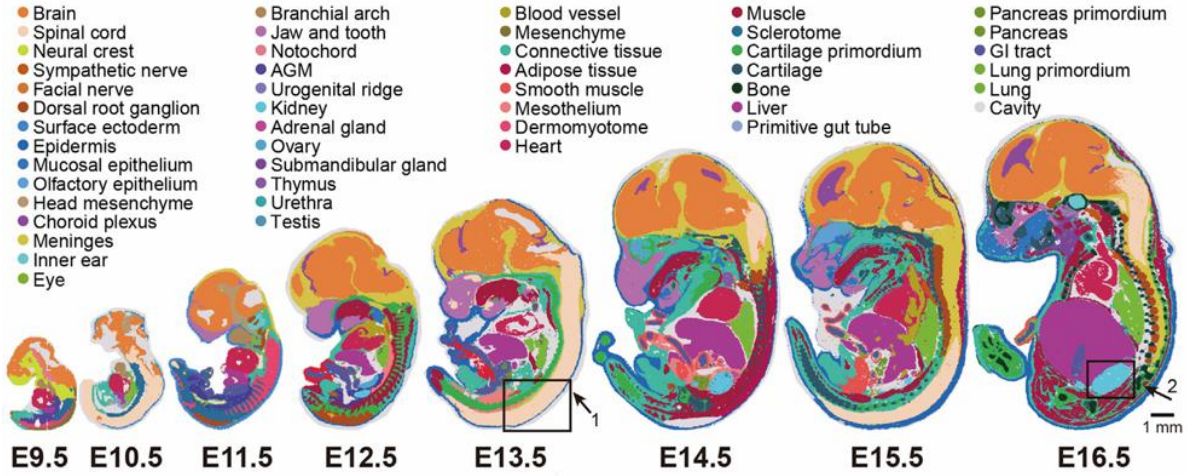


Wei, X, et al. Science 2022



Peng, Zhou, ... Nature Methods 2026

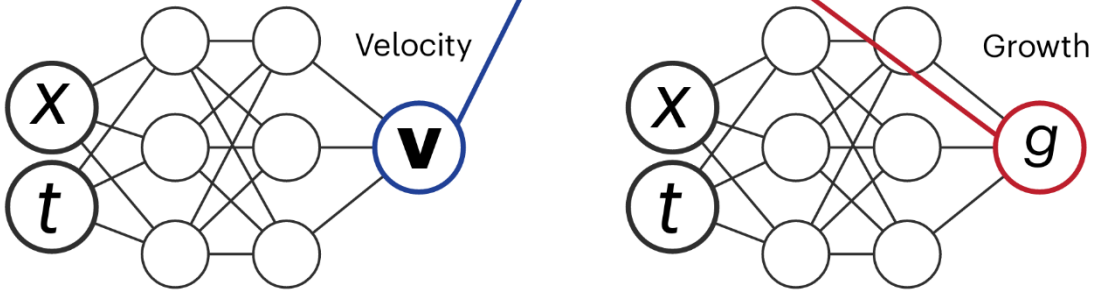
Dynamical systems modeling

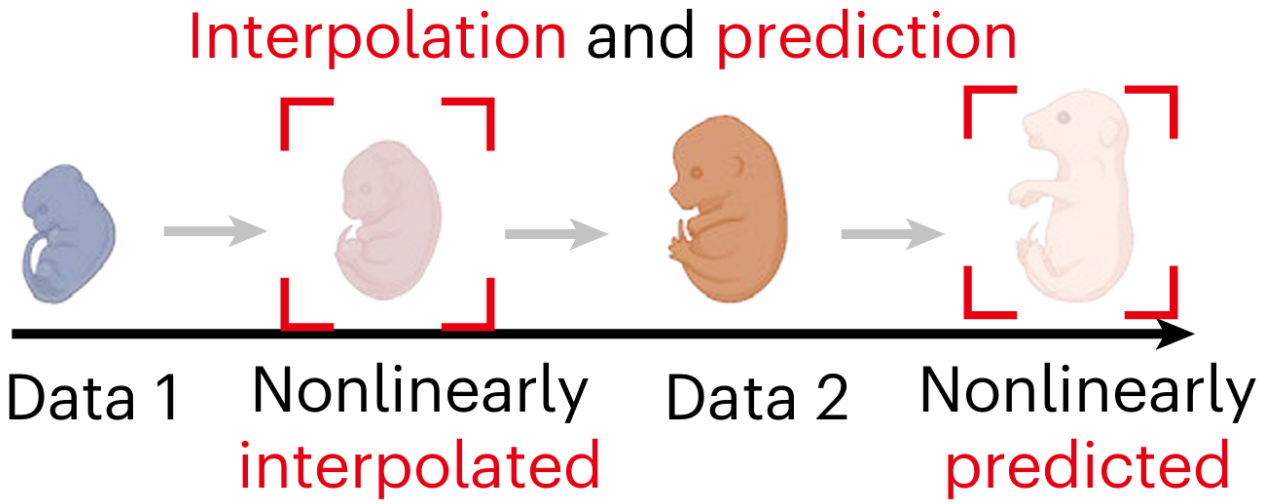


$$\frac{d}{dt}\begin{pmatrix}\boldsymbol{\phi}_t(\boldsymbol{x})\\ \ln m_t(\boldsymbol{x})\end{pmatrix} = \boxed{\begin{pmatrix}\boldsymbol{u}_t(\boldsymbol{\phi}_t(\boldsymbol{x}))\\ g_t(\boldsymbol{\phi}_t(\boldsymbol{x}))\end{pmatrix}}$$

$$\partial_t \rho_t(\boldsymbol{x}) + \nabla_{\boldsymbol{x}} \cdot (\boldsymbol{u}_t(\boldsymbol{x})\rho_t(\boldsymbol{x})) = g_t(\boldsymbol{x})\rho_t(\boldsymbol{x})$$

$$\partial_t \rho + \nabla \cdot (\mathbf{v}\rho) = g\rho$$





Optimal Transport: coupling between unpaired cells

$$\mathrm{WFR}_\delta^2(\mu_0,\mu_1)=\inf_{\rho,g,\boldsymbol{u}}\int_0^1\int_{\mathcal{X}}\frac{1}{2}(\|\boldsymbol{u}(\boldsymbol{x},t)\|_2^2+\delta^2\|g(\boldsymbol{x},t)\|_2^2)\rho_t(\boldsymbol{x})\mathrm{d}\boldsymbol{x}\mathrm{d}t$$

$$\text{s.t.}\qquad \partial_t\rho+\nabla_{\boldsymbol{x}}\cdot(\rho\boldsymbol{u})=\rho g,\ \rho_0=\mu_0,\ \rho_1=\mu_1,$$

traveling Dirac Liero et al. Inventiones mathematicae 2018

$$\mathrm{WFR\text{-}DD}_\delta^2(m_0\delta_{\boldsymbol{x}_0},m_1\delta_{\boldsymbol{x}_1})=2\delta^2(m_0+m_1-2\sqrt{m_0m_1}\overline{\cos}(\frac{\|\boldsymbol{x}_0-\boldsymbol{x}_1\|_2}{2\delta}))$$

$$m(t)=At^2-2Bt+m_0,\quad \boldsymbol{u}(t)m(t)=\boldsymbol{\omega}_0$$

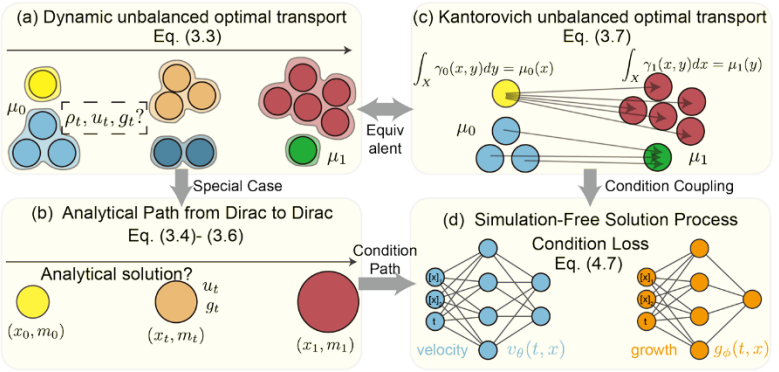
$$\begin{cases}A=m_0+m_1-2\sqrt{\dfrac{m_0m_1}{1+\tau^2}}\ ,\ B=m_0-\sqrt{\dfrac{m_0m_1}{1+\tau^2}}\\ \boldsymbol{\omega}_0=2\delta\tau\sqrt{\dfrac{m_0m_1}{1+\tau^2}}\boldsymbol{l}\ ,\ \tau=\tan(\dfrac{\|\boldsymbol{x}_1-\boldsymbol{x}_0\|_2}{2\delta})\end{cases}$$

Peng, Zhou, ... Nature Methods 2026 Peng, …, Zhou. ICLR 2026 Peng*, **Chen***, Zhou. NeurIPS 2026

Peng, Zhou, ... Nature Methods 2026 Peng, …, Zhou. ICLR 2026 Peng*, **Chen***, Zhou. NeurIPS 2026

$$\mathrm{WFR}_\delta^2(\mu_0,\mu_1) = \inf_{\rho,g,\boldsymbol{u}} \int_0^1 \int_{\mathcal{X}} \frac{1}{2}(\|\boldsymbol{u}(\boldsymbol{x},t)\|_2^2 + \delta^2\|g(\boldsymbol{x},t)\|_2^2)\rho_t(\boldsymbol{x})\mathrm{d}\boldsymbol{x}\mathrm{d}t$$

$$\text{s.t.} \qquad \partial_t\rho + \nabla_{\boldsymbol{x}}\cdot(\rho\boldsymbol{u}) = \rho g,\ \rho_0 = \mu_0,\ \rho_1 = \mu_1,$$



equivalent static Kantorovich

$$\mathrm{WFR}_\delta^2(\mu_0,\mu_1) = \inf_{(\gamma_0,\gamma_1)\in(\mathcal{M}_+(\mathcal{X}^2))^2} \int_{\mathcal{X}^2} \mathrm{WFR\text{-}DD}_\delta^2(\gamma_0(\boldsymbol{x},\boldsymbol{y})\delta_{\boldsymbol{x}},\gamma_1(\boldsymbol{x},\boldsymbol{y})\delta_{\boldsymbol{y}})\mathrm{d}\boldsymbol{x}\mathrm{d}\boldsymbol{y},$$

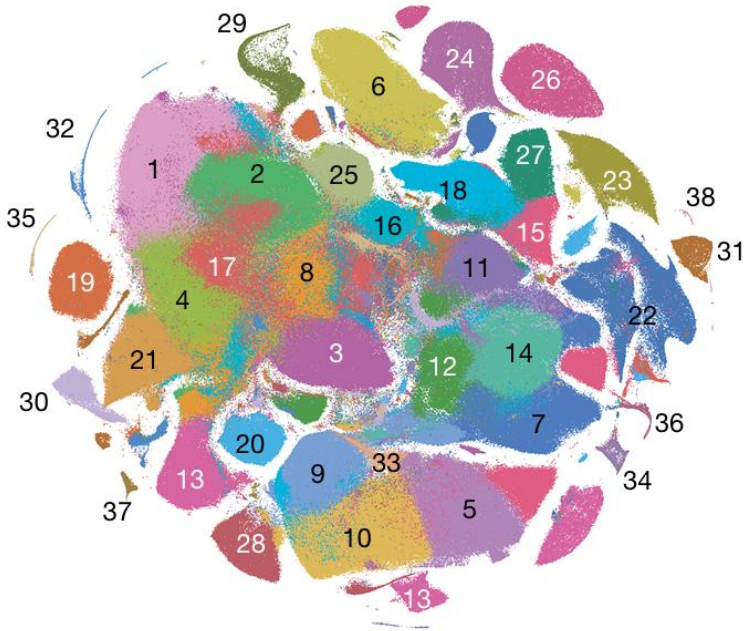
$$\Gamma(\mu_0,\mu_1) \overset{\text{def.}}{=} \left\{(\gamma_0,\gamma_1)\in\left(\mathcal{M}_+(\mathcal{X}^2)\right)^2 : \int_{\mathcal{X}}\gamma_0(\boldsymbol{x},\boldsymbol{y})\mathrm{d}\boldsymbol{y} = \mu_0(\boldsymbol{x}), \int_{\mathcal{X}}\gamma_1(\boldsymbol{x},\boldsymbol{y})\mathrm{d}\boldsymbol{x} = \mu_1(\boldsymbol{y})\right\}.$$

optimal entropy-transport (OET)

$$\mathrm{WFR}_\delta^2(\mu_0,\mu_1) = 2\delta^2 \inf_{\gamma\in\mathcal{M}_+(\mathcal{X}^2)}\{\int_{\mathcal{X}^2} -2\ln\overline{\cos}(\frac{\|\boldsymbol{x}-\boldsymbol{y}\|_2}{2\delta})\gamma(\boldsymbol{x},\boldsymbol{y})\mathrm{d}\boldsymbol{x}\mathrm{d}\boldsymbol{y}$$
$$+\mathrm{KL}(\int_{\mathcal{X}}\gamma(\boldsymbol{x},\boldsymbol{y})\mathrm{d}\boldsymbol{y}\|\mu_0(\boldsymbol{x})) + \mathrm{KL}(\int_{\mathcal{X}}\gamma(\boldsymbol{x},\boldsymbol{y})\mathrm{d}\boldsymbol{x}\|\mu_1(\boldsymbol{y}))\}$$

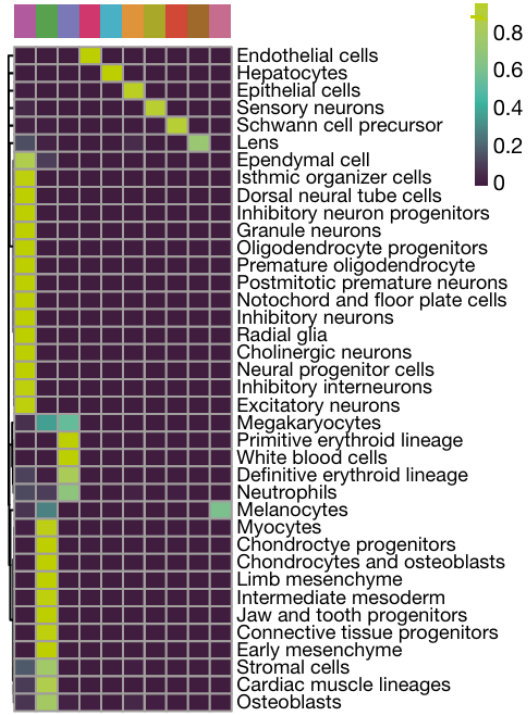
$$\gamma_0(\boldsymbol{x},\boldsymbol{y}) = \frac{\gamma(\boldsymbol{x},\boldsymbol{y})}{\int_{\mathcal{X}}\gamma(\boldsymbol{x},\boldsymbol{z})\mathrm{d}\boldsymbol{z}}\mu_0(\boldsymbol{x})$$
$$\gamma_1(\boldsymbol{x},\boldsymbol{y}) = \frac{\gamma(\boldsymbol{x},\boldsymbol{y})}{\int_{\mathcal{X}}\gamma(\boldsymbol{z},\boldsymbol{y})\mathrm{d}\boldsymbol{z}}\mu_1(\boldsymbol{y})$$

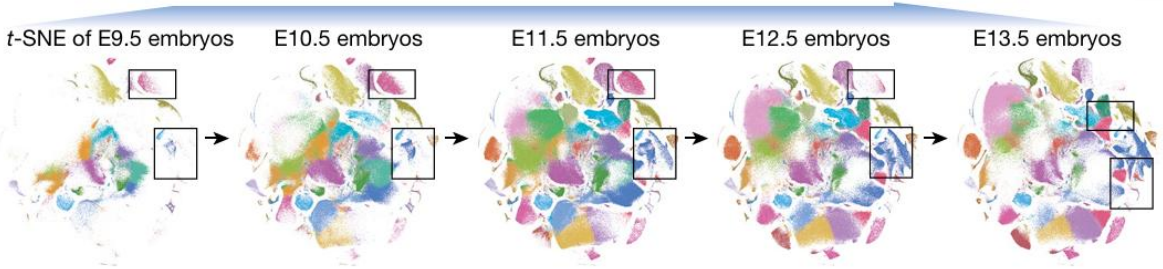
Toward atlas-scale single-cell datasets



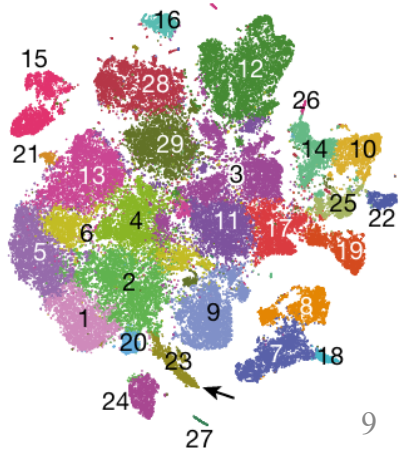
MOCA dataset annotation:

- Cell number: 1.3 million cells
- Major cell type: 38
- Sub cell type: 655



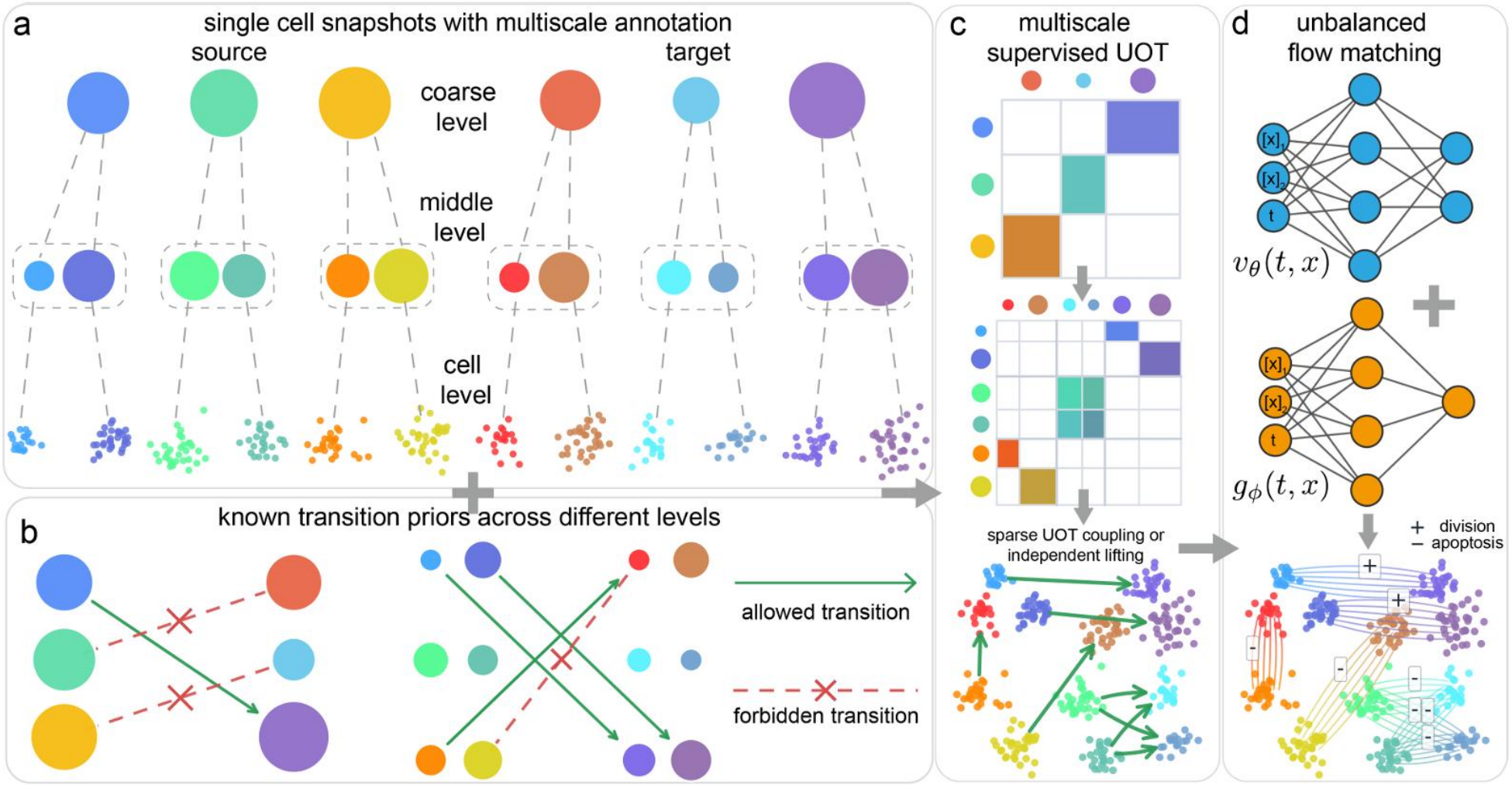


1–Keratinocyte (*Col19a1*$^+$)
2–Epidermal stem cells (*Meis1*$^+$)
3–Branchial arch ectodermal cells
4–Epidermal progenitors (*Igfbp2*$^+$)
5–Keratinocyte (*Brinp1*$^+$)
6–Keratinocyte (*Krt1*$^+$)
7–Otic vesicle epithelium
8–Otic sensory epithelium
9–Pericardium
10–Intestinal epithelium
11–Second branchial arch epithelium
12–Olfactory epithelium
13–Hair follicle stem cell
14–Intestinal stem cells
15–Renal epithelium
16–Retina epithelium
17–First branchial arch epithelium
18–Utricle and saccule epithelium
19–Lung epithelium
20–Suface ectoderm
21–Epidermal stem cells (*Lgr6*$^+$)
22–Endocrine cells
23–Apical ectodermal ridge (AER)
24–Urothelium
25–Stomach epithelium
26–Intestine epithelium (*Car3*$^+$)
27–Primordial germ cells
28–Doublets
29–Doublets



Cao et al. Nature 2019

Multiscale Supervised Unbalanced Optimal Transport Flow Matching

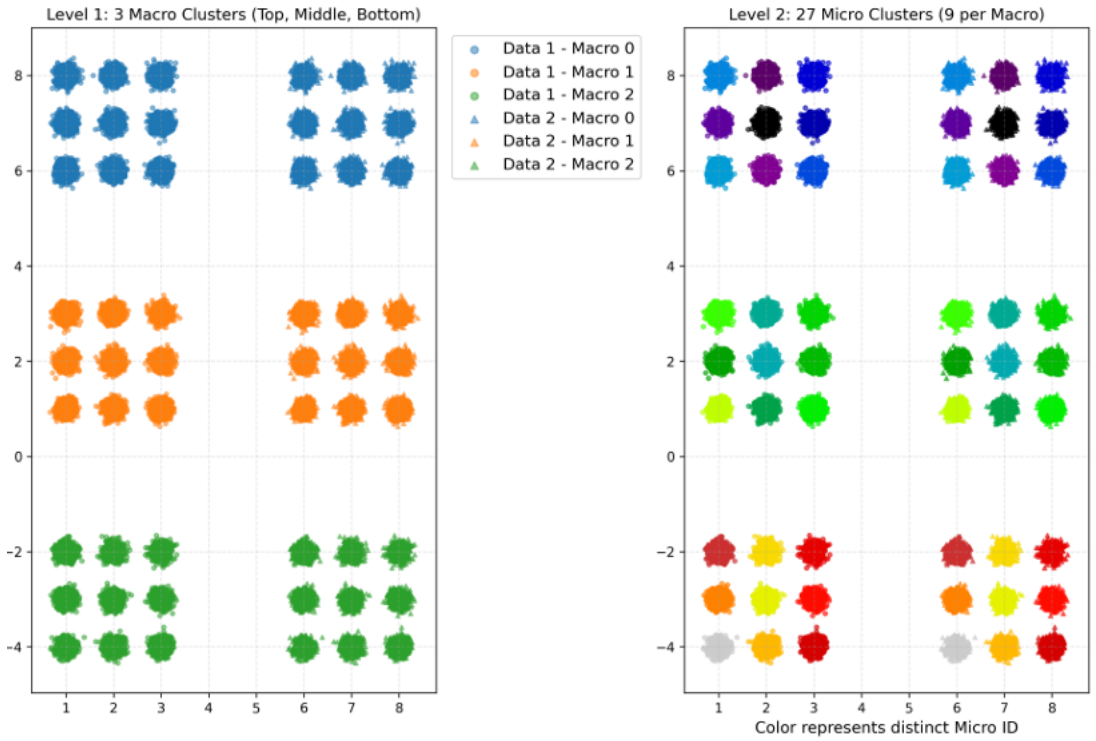


Peng*, **Chen***, Zhou. NeurIPS 2026

Peng*, **Chen***, Zhou. NeurIPS 2026

Table 1: **Coupling accuracy on the synthetic multiscale dataset (Q1).**

Method	Obj. Gap (%) ↓	Point Acc. ↑	Micro Acc. ↑	Macro Acc. ↑	Time (s) ↓
Ground Truth	0.00	1.000	1.000	1.000	–
Exact OT	2.35e-5	0.268	0.510	**1.000**	89.91
Mini-batch OT [33]	1.50e-1	0.002	0.328	0.988	8.86
Low-rank OT [31]	1.35e+1	0.000	0.076	0.695	151.92
MUST-FM (Sparse)	**1.78e-14**	**0.999**	**1.000**	**1.000**	10.50
MUST-FM (Independent)	3.99e-2	0.001	**1.000**	**1.000**	**0.025**



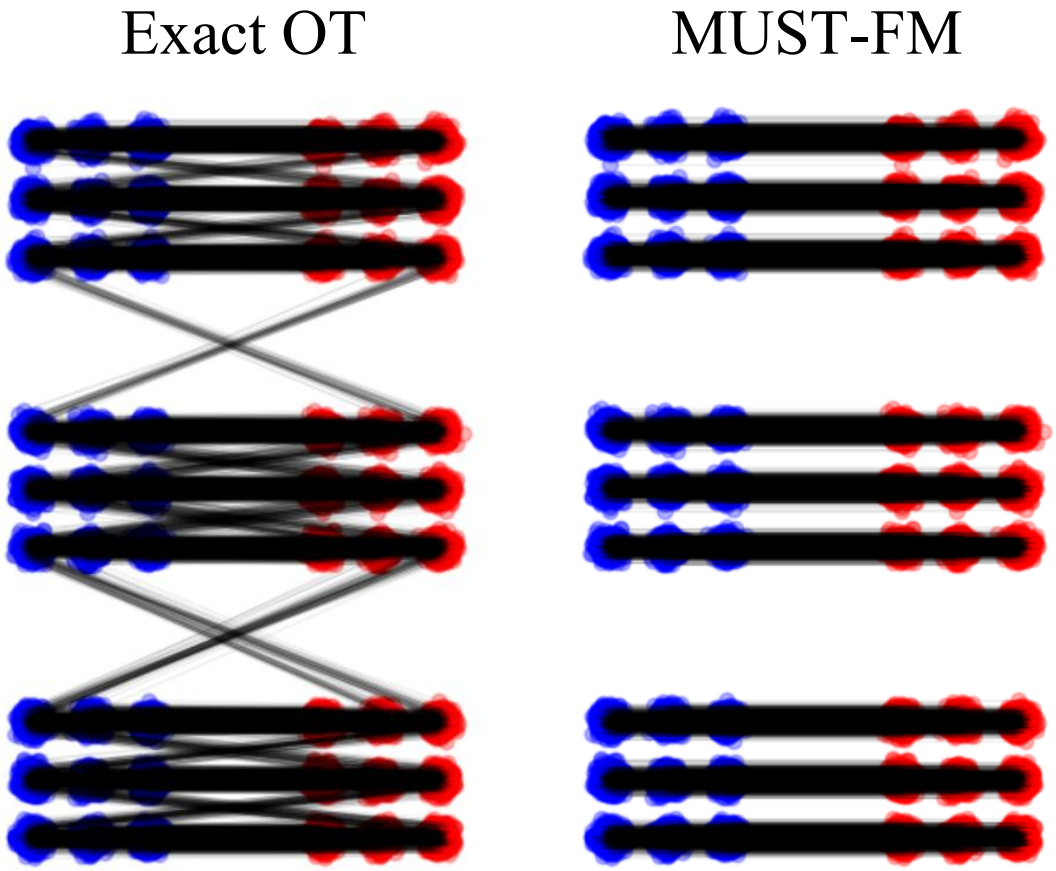


Table 3: **Leave-one-out interpolation on MOCA dataset [34] (Q3).** We report W_1 (↓) at each held-out time point.

Method	E10.5	E11.5	E12.5	Average
MUST-FM w/o Priors	1.370	1.429	**1.252**	1.350
MUST-FM w/ Priors	**1.326**	**1.364**	1.318	**1.336**

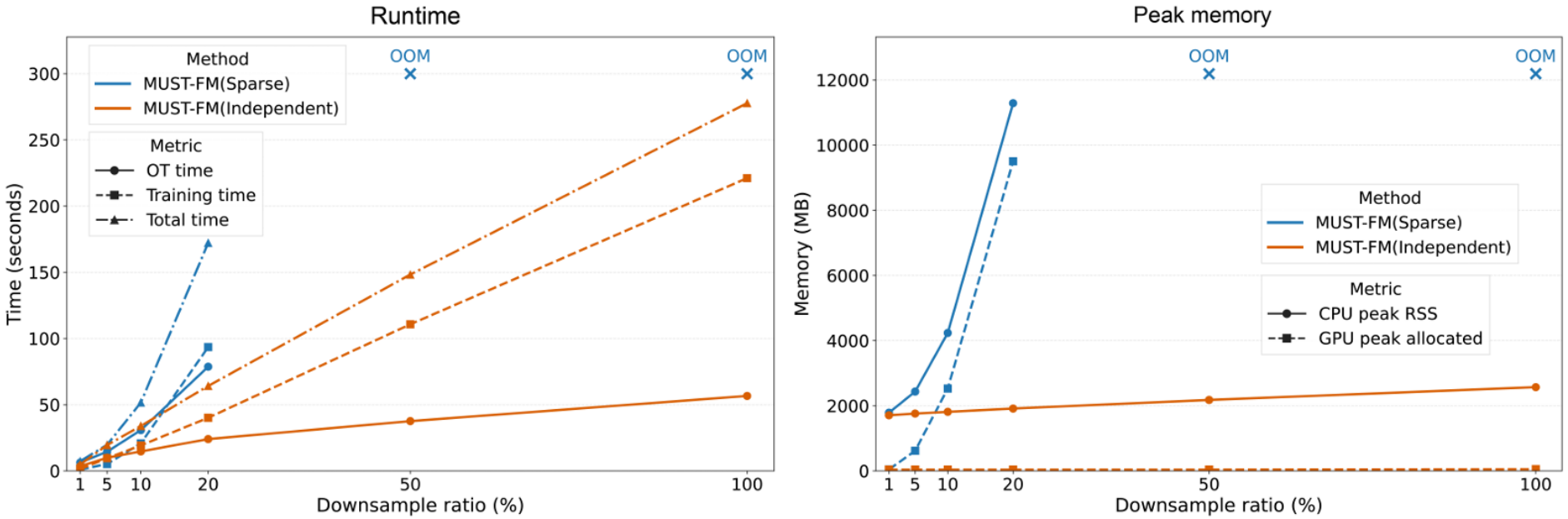
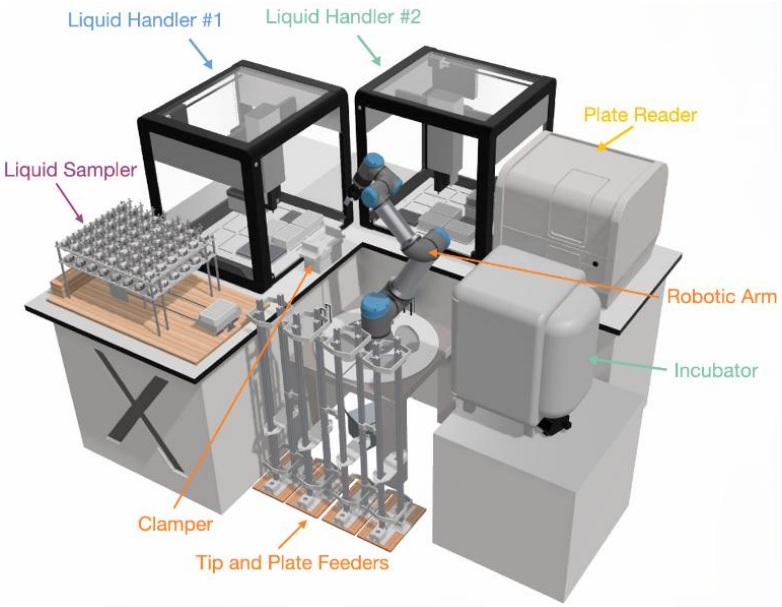


Figure 2: **Scalability on MOCA [34] dataset.** Runtime and memory usage of MUST-FM under different subsampling ratios.

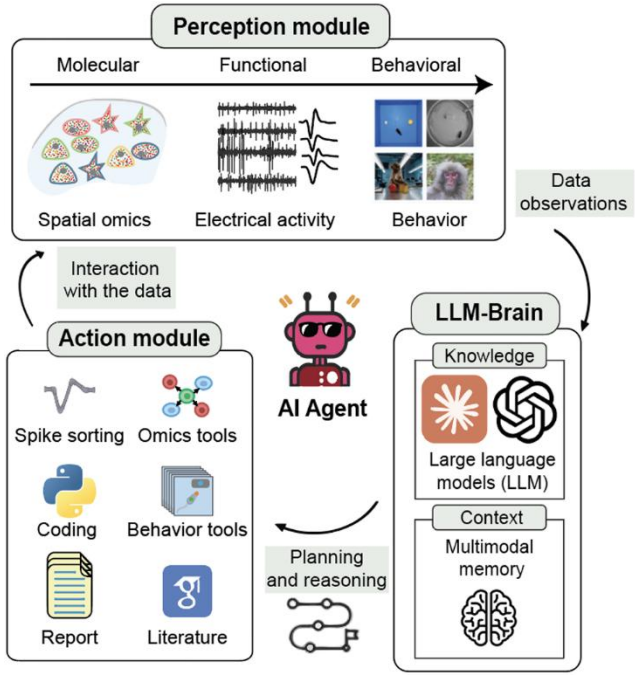
Future plan



Theoretical modeling

High-throughput data

Computational methods



Closed-loop validation